5:10 41%

CSA0720-MCQ

Total points 13/20

65%

CO3-AT3

7.A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is: *1/1

- a) Increase UDP buffer size
- b) Switch to TCP
- c) Implement Forward Error Correction (FEC) ✓
- d) Reduce frame rate to 1 fps

6.UDP is preferred over TCP for DNS queries primarily because: *0/1

- a) UDP supports fragmentation ✗

✓ 5.In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by: *1/1

- a) MTU size
- b) CWND growth rate relative to RTT ✓
- c) DNS resolution time
- d) UDP competition

X 11.Head-of-line blocking in HTTP/1.1 occurs when: *0/1

- a) The DNS server is slow X
- b) A response blocks subsequent responses on the same connection
- c) TCP CWND drops to zero
- d) The server rejects pipelined requests

Correct answer

Name of the student-Reg.No *

Shaik.imran(192525348)

✓ 13.Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks? *1/1

- (●) a) 301 ✓
- ( ) b) 302
- ( ) c) 304
- ( ) d) 307

✓ 20.Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game? *1/1

- ( ) a) TCP for game state, HTTP for assets

18.Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases: *0/1

- a) Cache hit ratio
- b) Recursive resolver query load
- c) DNSSEC signature validity
- d) Zone transfer frequency

Correct answer

- b) Recursive resolver query load

14.In HTTP caching, the Cache-Control: no-cache directive means: *1/1

- a) Never cache this resource
- b) Validate with the server before using a cached copy
- c) Cache for 0 seconds
- d) Disable all caching headers

5:11 41%

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8.What is the maximum UDP payload size without IP-level fragmentation on a standard Ethernet link (MTU = 1500 bytes)? *1/1

- a) 1480 bytes
- b) 1472 bytes
- c) 1460 bytes
- d) 1500 bytes

17.Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID? *1/1

- a) ARP poisoning
- b) DNS cache poisoning
- c) SYN flooding
- d) BGP hijacking

connection overhead is unnecessary

c) TCP cannot use port 53

d) UDP has better error correction

Correct answer

b) DNS responses are small and connection overhead is unnecessary

12.HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at the: *0/1

a) Application layer

b) DNS layer

c) TCP layer

d) IP layer

Correct answer

c) TCP layer

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a) TCP for game state, HTTP for assets

b) UDP with application-layer sequencing for game state, HTTP/3 for assets ✓

c) HTTP/2 for all traffic

d) TCP with Nagle enabled for all traffic

✓ 3.Which field in the TCP header is used to implement flow control at the receiver side? *1/1

a) Sequence number

b) Acknowledgement number

c) Window size ✓

d) Urgent pointer

X 2.A TCP sender has CWND = 8 MSS and SSTHRESH = 16. After one RTT with no loss, CWND becomes: *0/1

✓ 10.HTTP/1.1 persistent connections reduce latency by: *1/1

- ○ a) Enabling UDP fallback
- ◉ b) Reusing the same TCP connection for multiple requests ✓
- ○ c) Compressing HTTP headers
- ○ d) Sending requests without waiting for DNS

✓ 1.Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs? *1/1

- ○ a) Slow Start
- ○ b) Fast Retransmit
- ◉ c) Fast Recovery ✓
- ○ d) Tahoe Reset